

Question				Marking details	Marks available						
					AO1	AO2	AO3	Total	Maths	Prac	
11.	(a)			Empirical formula percentage oxygen = 35.6 C : H : O = $\frac{53.3}{12} : \frac{11.1}{1} : \frac{35.6}{16}$ (1) = 4.44 : 11.1 : 2.22 = 4 : 10 : 2 $\Rightarrow$ C₂H₅O (1) Mass spectrum <i>M</i>_r = 90 (1) molecular formula C₄H₁₀O₂ (1) any fragment linked to m/z (1) IR O—H peak at 3200-3550 (1) NMR 3 hydrogen environments (1) Potassium dichromate(VI) X is 1° alcohol, 2° alcohol or aldehyde (1) Sodium carbonate (carboxylic) acid not formed so X must be 2° alcohol / oxidised to ketone (1) X is butane-2,3-diol / unambiguous formula (1)	1	1 1				1 1	

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	(b)	(i)		redox / oxidation	1			1		1
		(ii)		$\text{H}_3\text{C}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{CH}_3$ ecf possible from part (a) e.g. if 1° alcohol given			1	1		
				Question 11 total	2	7	3	12	1	3